MA3K7 Assignment 5 Rubric

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Question: Conquer the Tile Game

The integers $1, 2, \ldots, n$ are placed on the vertices of a spiral grid, with 1 at the centre. Each vertex is labelled by its number.

Two players play the game on the same board. Each player has their own position, both starting at the centre, which is already unlocked and worth 0 points.

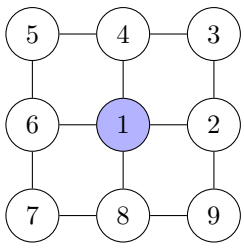
Players take turns. On each turn, a player chooses a vertex adjacent to their current position.

- If the chosen vertex is already unlocked, the player moves to that vertex.
- If the chosen vertex is locked and labelled m, the player attempts to unlock it. The first attempt to unlock that vertex succeeds with probability $1/m$. Each failed attempt increases the success probability for future attempts to unlock that vertex by $1/m$, up to a maximum of 1. This probability depends only on the number of previous failed attempts to unlock that vertex, regardless of which player made them.
  - If the attempt succeeds, the vertex becomes unlocked, the player moves to that vertex, and scores m points (provided it has not already been scored).
  - If the attempt fails, the player remains at their current position.

Each vertex can be scored at most once and once a vertex has been scored by either player, it is worth 0 points thereafter.

The game ends when every vertex has been scored. The player with the higher total score wins. Investigate the game, in particular whether the player who first attempts a vertex has an advantage.

Below is an example of the board when $n = 9$.



1 Entry

I know

From the rules of the game, I know that

- the board is a spiral grid with vertices labelled $1, 2, \ldots, n$;
- each vertex can be scored at most once;
- if a locked vertex has label m, then successive attempts to unlock it succeed with probabilities
$$\frac{1}{m}, \frac{2}{m}, \ldots, \frac{m}{m};$$

- once a vertex is eventually unlocked, exactly m points are awarded to the player who unlocks it first.

TRY

My first thought was, to look only at the first attempt on a vertex m. That gives expected score

$$\frac{1}{m} \cdot m = 1.$$

But that only describes a single attempt, not the whole game.

AHA

In the full game, attempts continue until success. Since the success probability eventually reaches 1, every vertex is guaranteed to be unlocked at some point. So the total score contributed by vertex m is always exactly m.

Therefore

$$S_A + S_B = \sum_{m=2}^{n} m,$$

where S_A and S_B are the total scores of the two players.

NOTE

So the randomness, does not change the total amount of score in the game. It only changes which player receives each labelled vertex.

NOTATION

For brevity, I use the following notation.

- Let $X_m \in \{A, B\}$ denote the player who scores vertex m.
- For a fixed vertex labelled m, let r be the number of previous failed attempts to unlock that vertex.
- Let
$$p_r = \frac{r+1}{m}, \qquad 0 \le r \le m-1,$$
be the success probability on the next attempt after r failures.
- Let q_r be the probability that the player whose turn it is eventually scores that vertex, given that there have already been r failures.

Then

$$S_A = \sum_{m:X_m=A} m, \qquad S_B = \sum_{m:X_m=B} m.$$

I WANT

What I want to understand, is first whether the player who starts the contest at a single vertex has an advantage, and then how that local race interacts with movement on the spiral.

LOCALISE

To start somewhere simpler, I first isolate a single vertex and ignore movement.

2 Attack

2.1 A single contested vertex

Fix a vertex labelled m, and suppose both players keep contesting this same vertex.

If there have already been r failures, then on the current turn

- with probability $p_r = \frac{r+1}{m}$, the current player succeeds immediately;
- with probability $1 - p_r$, the current player fails, and then the opponent faces the same situation with $r + 1$ failures.

So

$$q_r = p_r + (1 - p_r)(1 - q_{r+1}).$$

Also, when $r = m - 1$, the next attempt succeeds for sure, so

$$q_{m-1} = 1.$$

2.2 Closed form

TRY The backward recursion is easy to compute numerically, but I wanted a cleaner formula.

Define

$$s_j := q_{m-1-j}, \qquad 0 \le j \le m-1.$$

Then

$$s_0 = 1,$$

and the recurrence becomes

$$s_j = 1 - \frac{j}{m} s_{j-1} \qquad (1 \le j \le m-1).$$

Proposition 2.1. *For each* $0 \le j \le m-1$,

$$s_j = \sum_{k=0}^{j} (-1)^k \frac{j!}{(j-k)!\, m^k}.$$

In particular,

$$q_0(m) = \sum_{k=0}^{m-1} (-1)^k \frac{(m-1)!}{(m-1-k)!\, m^k}.$$

Proof. I prove the formula by induction on j.

For $j = 0$, we have $s_0 = 1$, which matches the formula.

Now suppose the formula holds for $j-1$. Then

$$s_j = 1 - \frac{j}{m} s_{j-1} = 1 - \frac{j}{m} \sum_{k=0}^{j-1} (-1)^k \frac{(j-1)!}{(j-1-k)!\, m^k}.$$

Since

$$\frac{j}{m} \cdot \frac{(j-1)!}{(j-1-k)!\, m^k} = \frac{j!}{(j-1-k)!\, m^{k+1}},$$

we get

$$s_j = 1 - \sum_{k=0}^{j-1} (-1)^k \frac{j!}{(j-1-k)!\, m^{k+1}}.$$

Re-indexing with $\ell = k+1$ gives

$$s_j = 1 - \sum_{\ell=1}^{j} (-1)^{\ell-1} \frac{j!}{(j-\ell)!\, m^\ell} = 1 + \sum_{\ell=1}^{j} (-1)^\ell \frac{j!}{(j-\ell)!\, m^\ell}.$$

So

$$s_j = \sum_{\ell=0}^{j} (-1)^\ell \frac{j!}{(j-\ell)!\, m^\ell},$$

as required. □

2.3 Small values

TRY I first computed the first few values exactly, to see what the local race looks like.

$m = 2$.

$$q_1 = 1, \qquad q_0 = \frac{1}{2}.$$

$m = 3$.

$$q_2 = 1, \qquad q_1 = \frac{2}{3}, \qquad q_0 = \frac{1}{3} + \frac{2}{3}\left(1 - \frac{2}{3}\right) = \frac{5}{9}.$$

$m = 4$.

$$q_3 = 1, \qquad q_2 = \frac{3}{4}, \qquad q_1 = \frac{1}{2} + \frac{1}{2}\left(1 - \frac{3}{4}\right) = \frac{5}{8},$$

$$q_0 = \frac{1}{4} + \frac{3}{4}\left(1 - \frac{5}{8}\right) = \frac{17}{32}.$$

I FIND

From these first cases, I get

$$q_0(2) = \frac{1}{2}, \qquad q_0(3) = \frac{5}{9}, \qquad q_0(4) = \frac{17}{32}.$$

So from $m = 3$ onward, the first attempter does better than $\frac{1}{2}$ in the cases I checked exactly.

2.4 Python check

INTRODUCE

I also computed $q_0(m)$ exactly for larger m, using Python and rational arithmetic.

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# code in github / Question5
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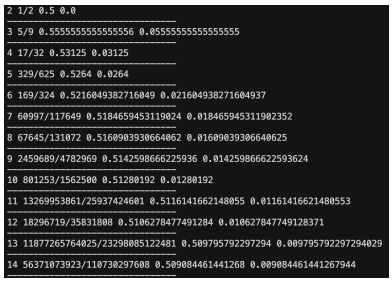

Figure 1: Output result (m, $q_0(m)$, and $q_0(m) - 0.5$)

OBSERVATION

The computed values, stay above $\frac{1}{2}$ in every case I tested, but the excess seems to shrink as m grows.

This led me to consider

$$\Delta(m) := m\left(q_0(m) - \frac{1}{2}\right),$$

which measures the size of the first-move premium at a vertex of value m.

2.5 Behaviour of the premium

CHECK

From the computed values, $\Delta(m)$ appears to remain bounded as m grows, and in the examples I checked it stays quite small.

So even when m is large, the extra benefit of being the player who starts the local race does not seem to grow proportionally with the label of the vertex.

I do not have a completely clean proof of this stronger statement from the recurrence alone, so I treat it here as a checked numerical pattern rather than as a separate theorem.

2.6 Back to the spiral

RE-ENTER

Now I return to the actual board, and try to see what the local calculation says about the full game.

On the $n = 9$ spiral, the centre is adjacent only to

$$2, 4, 6, 8.$$

So the larger labels $3, 5, 7, 9$ are not available immediately. They must be reached through smaller vertices:

$$\begin{array}{ll} 3 \text{ from } 2 \text{ or } 4, & 5 \text{ from } 4 \text{ or } 6, \\ 7 \text{ from } 6 \text{ or } 8, & 9 \text{ from } 2 \text{ or } 8. \end{array}$$

AHA

So an early move, can matter for two different reasons:

- it may score the current vertex;
- it may control access to a larger vertex later.

This means the game is not only about the value written on the current square. It is also about which future vertices that move opens up.

STUCK

At this point, I do not see a full optimal strategy for the entire spiral. The local model is clean because both players keep contesting one vertex, but in the real game a player can switch targets, move through already-open vertices, or avoid a race completely.

So I cannot claim a complete global solution.

CONJECTURE

This leads me, to the following conjecture.

Conjecture 1. *On the full spiral, positional control should matter more than the local first-attempt advantage once the labels are moderately large. In particular, good play should prioritise vertices that give access to valuable future vertices, not just the largest label currently available.*

3 Solution

3.1 What is proved exactly

CLAIM

For a single contested vertex labelled m, the local race is described exactly by Proposition 2.1:

$$q_0(m) = \sum_{k=0}^{m-1} (-1)^k \frac{(m-1)!}{(m-1-k)!\, m^k}.$$

The first few exact values are

$$q_0(2) = \frac{1}{2}, \qquad q_0(3) = \frac{5}{9}, \qquad q_0(4) = \frac{17}{32},$$

and the Python computation shows the same pattern for larger tested values.

CHECK

From the exact small cases and the numerical data, the player who first attempts a single vertex does seem to have an advantage. What I have proved exactly is the formula for that local probability. What I have checked computationally is that this advantage persists, while appearing to become relatively small for large m.

3.2 Why positional control seems more important

COMPARE

Suppose a vertex of value m is contested, and I reach it first. My expected score from that vertex is

$$mq_0(m).$$

If instead I can reach that vertex without a race, then I expect close to m points.

So the gain from avoiding a contest at that vertex is

$$m - mq_0(m) = m(1 - q_0(m)).$$

For large vertices, the numerical behaviour of $q_0(m)$ suggests that this positional effect can be much larger than the local first-move premium.

CASE

On the $n = 9$ board, the clearest example is vertex 9.

If both players can approach it, then there is a local race for those 9 points. But if one player controls the approach through 2 or 8 and the other does not, then the more important issue is not who gets the first try at 9, but who has access to 9 at all.

The same picture appears, in weaker form, for $3, 5, 7$.

3.3 Answer to the original question

CHECK

The original question asks, in particular, whether the player who first attempts a vertex has an advantage.

For a single contested vertex, my calculations and computations strongly suggest that the answer is yes, and the exact local probability is given by Proposition 2.1.

For the full spiral, I do not have a complete theorem. What my work suggests instead is that the local first-attempt effect is usually smaller than the positional effect of controlling access to valuable vertices.

So the best summary I can give is that there appears to be a local first-player advantage, but the full game is governed just as much by movement and access as by the local unlocking probabilities.

4 Review

4.1 Key moments

REFLECT

At the beginning, I was trying to understand the whole game at once, but that quickly became too messy because movement and probability interact.

The first useful step was to separate the problem into a local question and a global question. Once I fixed a single vertex and wrote down the recurrence for q_r, the problem became much more manageable.

A second useful step was the substitution

$$s_j = q_{m-1-j},$$

which turned the backward recursion into a cleaner forward induction.

4.2 Mistakes and corrections

MISTAKE

My first instinct, was that a large label should come with a large first-player advantage. That turned out not to be the right way to think about the game.

The computations suggest that the extra benefit of moving first stays fairly small, even when the label m is large.

AHA

This changed the picture for me, because the more important question became not just who gets the first try, but who turns a valuable vertex into a contest in the first place.

4.3 What I have and have not solved

CHECK

I have solved the local race exactly, in the sense that I obtained a closed formula for $q_0(m)$.

I have not solved the full spiral game optimally. The global discussion in this rubric should therefore be read as a structural conclusion and a conjectural strategy picture, not as a complete theorem.

4.4 Extensions

EXTEND

One extension which was in fact the original idea I had in mind, is to consider a variant with a shared token rather than two independent player positions.

In that version, each move would also determine the opponent's next position. So moving towards a valuable vertex would simultaneously bring the opponent closer to it. This creates a much stronger positional interaction, where access and timing cannot be separated as in the current model.

The analysis in the current version does not directly apply to that setting, since the local model assumed that players can repeatedly contest a single vertex independently of movement. Understanding how the local race interacts with forced movement in the shared token version would likely require a different approach.

Another possible extension would be to modify the unlocking rule, for example by increasing the success probability by $\frac{c}{m}$ at each attempt instead of $\frac{1}{m}$, and investigating how this changes the size of the first-player advantage.

Finally, the global behaviour could be explored computationally. For small values of n, one could run an exact decision tree search to test whether the positional control picture suggested here is reflected in optimal play.

Code

The code for this assignment can be found on my GitHub page:
`https://github.com/cathal0317/Problem_Solving_RUBRIC`